Intelligent Perimeter Security System with Reduced False Alarms Using Sensor Fusion

Review 1

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ABSTRACT

This project presents a prototype smart intrusion detection and deterrent system designed to reduce false alarms common in traditional security setups. A PIR sensor and an ultrasonic sensor work together as a dual-trigger layer, requiring both to detect activity before proceeding — filtering out false triggers early. Once confirmed, an ESP32-CAM with an onboard AI/ML model captures and analyzes the scene to verify if the activity is a genuine intrusion. If confirmed, the system activates a non-harmful deterrent, such as light or sound, to discourage the intruder safely. This single-unit prototype demonstrates a low-cost, intelligent, and ethical approach to smart security.

PROJECT OBJECTIVES

- To design and demonstrate a low-cost, single-unit intrusion detection and deterrence system that addresses the persistent problem of high false alarm rates observed across existing commercial and research-based security systems.
- To reduce false triggers using a dual-sensor confirmation approach (PIR + ultrasonic sensor), requiring agreement between both sensors before escalating to further action — rather than relying on a single sensor.
- To introduce an intelligent confirmation layer using an ESP32-CAM with an AI/ML-based classification model, activated only after the dual-sensor trigger confirms potential activity — reducing unnecessary computation and power draw by keeping the camera/ML system dormant until needed.
- To explore whether meaningful person/object classification can be achieved on constrained edge hardware (ESP32-CAM).
- To respond to confirmed intrusions using a strictly non-harmful deterrent system (light and/or sound and/or sprinkler), in contrast to systems that rely on aggressive or incapacitating countermeasures, prioritizing safety and ethical responsibility over aggressive deterrence.
- To build and validate a working proof-of-concept hardware prototype as a demonstration of the proposed detection-confirmation-deterrence pipeline, establishing feasibility for future expansion (e.g., multi-node deployment).
- To keep the system self-contained and independent of cloud infrastructure for its core detection and deterrent functions, ensuring the device operates reliably as a standalone unit.

LITERATURE REVIEW

S. No.	Title of Manuscript	Publisher Name	Year of Publishing	Observations/ Work done in Manuscript
1.	Cross border intruder detection in hilly terrain in dark environment	Elsevier	January 2016	• The system uses Microsoft Kinect for automatic border surveillance, through skeletal data collected. • Hidden Markov Model (HMM) is employed to classify human actions, making the recognition process more robust and reliable. • Uses the skeletal data obtained from Kinect to detect whether the person is standing, walking, bending,crawling • Able to differentiate between human and animal,triggers alarm if its human • Kinect relies on IR-based depth sensing rather than visible light, eliminates the bottleneck of poor illumination • The depth approximation feature triggers alarm only when it detects human movement crossing a certain threshold, thereby reducing false detections. • The proposed method achieves an overall detection accuracy of 92%, significantly reducing nuisance alarms and allowing security personnel to respond only to genuine intrusions. • The system is robust to noisy Kinect data, works for new users without additional training, and can be extended to recognize more human actions in the future.

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2.	Intelligent intrusion detection system featuring a virtual fence, active intruder detection, classification, tracking, and action recognition	Elsevier	2018	• Image-based Intelligent Intrusion Detection System (IIDS) with virtual fencing, real-time detection, object classification, tracking, and behavior recognition, eliminating the need for continuous human monitoring. • Integrated optical and thermal cameras improve intrusion detection in low-light, adverse weather, and camouflaged conditions, overcoming the limitations of conventional optical cameras • virtual function has function to set boundary between surveilliance and external zones in GUI to define a early pre-alarm if necessary • background model to detect objects with network model based on CNN to classify objects as either intruder or wild animals. • Real-time alarms are triggered when an intruder crosses the virtual fence, and an optimized LRCN model recognizes five intrusion behaviors: walking, running, crawling, climbing, and jumping. • Functions as autonomous surveillance sensor, reducing reliance on human operators while improving the reliability and efficiency

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3.	Detection of an Intruder and Prediction of His State of Motion by Using Seismic Sensor	IEE	2017	• Uses geophone sensors to capture seismic vibrations (footfall data) generated by human footsteps for reliable intruder detection. Since geophones are buried underground, they are difficult for intruders to detect or tamper with. • Proposes an adaptive event extraction technique that accurately detects individual footfall events and extracts only the relevant portion of the seismic signal. This reduces noise and improves feature extraction for machine learning. • Employs an SVM-RBF (Support Vector Machine with Radial Basis Function kernel) classifier to identify the presence of an intruder and classify their state of motion (walking, jogging, or running) using extracted seismic features. • Achieves an intruder detection accuracy of up to 86% using only 2 seconds of seismic data, while predicting the intruder's state of motion with 77% accuracy using 6 seconds of data.

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				• PIR (Passive Infrared) sensors are less suitable because they have a limited coverage area (about 10 m with a 110° field of view) and cannot be effectively camouflaged, making them easier for intruders to identify and avoid. • Seismic sensors can distinguish between different types of movement, such as human footsteps vs. quadrupeds (animals), and have also been used to differentiate male and female footsteps with high accuracy. • Floor vibration sensing has been used for indoor identification of individuals by analyzing their footsteps. However, this approach performs well mainly in controlled indoor environments and is less suitable for outdoor surveillance. • A hybrid system combining PIR, infrared (IR), and seismic sensors can significantly improve intrusion detection performance, achieving approximately 97.2–97.3% accuracy for distinguishing humans from animals.

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4.	Empirical Study of Stepped FM UWB Sensor for Indoor Intruder Detection	John Wiley & Sons, Ltd	2015	• The Stepped-FM UWB (Ultra Wideband) based intruder detection system is designed to detect intruders, estimate their distance, and continuously track their movement inside a building. • Traditional camera and infrared (IR) security systems detect movement only within their Line of Sight (LOS), making them ineffective when obstacles block the view. In contrast, radio-wave based systems can perform Non-Line of Sight (NLOS) detection, allowing them to detect movement through certain obstacles. However, conventional radio wave systems cannot accurately determine the intruder's distance from the sensor. • Ultra-Wideband Impulse Radio (UWB-IR) sensors provide excellent multipath characteristics, enabling better distance estimation using reflected radio waves. However, they have several drawbacks: ○ Difficulty distinguishing between an actual intruder and a visitor approaching the building. ○ Require high-speed Analog-to-Digital (A/D) converters to capture ultra-short pulses.

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				○ Require highly precise synchronization (sync) circuits, increasing system complexity and cost. • The proposed Stepped-FM UWB system overcomes these limitations by transmitting narrowband signals with stepwise frequencies, eliminating the need for fast A/D converters while still achieving high distance resolution after signal processing. • The transmitting and receiving antennas are placed near the center of the room (monostatic configuration) to monitor surrounding changes, detect intruder movement, and estimate the corresponding distance. • The system detects an intruder by observing changes in the range spectrum. Due to the monostatic antenna configuration, the intruder's position can be localized using concentric circles around the antenna, allowing continuous tracking of the intruder's movement and distance. This also helps distinguish intruders from visitors, thereby reducing false detections.

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				• The monitored area is divided into multiple concentric radial zones, each associated with a different alarm level: ◦ Zone 1: Attention ◦ Zone 2: Danger ◦ Zone 3: Intrusion • As an intruder approaches or crosses these zones, the system generates progressively higher alert levels, enabling early warning and intrusion pre-detection.

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5.	IoT-Inspired Framework of Intruder Detection for Smart Home Security Systems	Web of Science	2020	• Smart foot-mat-based intruder monitoring and detection system for enhancing home security using IoT enabled sensors. • Uses Fog Computing to analyze identity-based parameters such as foot size, foot pressure, and movement in real time. • Classifies users into Authentic and Non-Authentic categories using the Bayesian Belief Model (BBM). • Applies the Temporal Granulation Process (TGP) to process identity data and generate the Authentic Index (AI) for identity prediction. • Employs the Adaptive Neuro-Fuzzy Inference System (ANFIS) to predict user authenticity and generate real-time intrusion alerts. • Validated using 49,695 smart home data instances, demonstrating its effectiveness in real-world home security applications. • Integrates IoT, Fog Computing, and Cloud Computing for efficient data collection, processing, and decision making.

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				• Consists of four phases: Data Perception, Data Analysis, Data Extraction, and Intelligent Prediction. • Provides reliable, efficient, and real time intrusion detection to improve home security.

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6.	An Advanced IoT Based Border Surveillance and Intrusion Detection System	IEEE	2023	• Proposes a low-cost IoT-based Border Surveillance and Intrusion Detection System (BSIDS) by integrating multiple sensors such as IR sensor, PIR sensor, TF Mini-S LiDAR sensor, seismic sensor, acoustic sensor, Arduino Uno, ArduCam, and an alarm system. The combination of these sensors enables continuous border monitoring by detecting motion, vibrations, sound, and object distance, improving intrusion detection compared to conventional surveillance systems. • Implements a real-time monitoring system using a multi-sensor network, where data collected from all sensors is analyzed and uploaded to a cloud server. The processed information can be accessed through web and desktop applications, and the system automatically sends instant alerts to the base station whenever suspicious movement or unauthorized border crossing is detected. • Integrates multiple sensing technologies into a single surveillance platform to improve detection accuracy, situational awareness, and response time. Unlike traditional systems that mainly rely on CCTV cameras or physical barriers, the proposed system combines infrared, motion, vibration, sound, and distance sensing to reduce human intervention and enhance border security.

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				• Provides a scalable and efficient surveillance architecture capable of 24/7 border monitoring with minimal manual effort. The cloud-based framework ensures reliable data storage, remote access, and seamless integration with existing border security infrastructure, making it suitable for monitoring large and remote border areas. • Suggests future enhancements by incorporating Artificial Intelligence (AI) for intelligent threat analysis, drones and autonomous vehicles for patrolling inaccessible regions, blockchain for secure and tamper-proof storage of surveillance data, and quantum computing for faster processing of sensor data. These technologies can significantly improve the accuracy, scalability, security, and overall performance of the proposed border surveillance system.

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7.	Border Security and surveillance System Using IOT	An Advanced IoT Based Border Surveillance and Intrusion Detection System	2022	• Proposes an IoT-based Border Security and Surveillance System that uses drones, surveillance cameras, and wireless IoT sensors for continuous 24/7 border monitoring. The system is designed to detect, track, and report intruders in real time, helping prevent terrorism, smuggling, illegal immigration, and other border threats. • Implements a drone-based surveillance approach, where drones equipped with cameras and sensors capture live video of the border area. The collected data is transmitted to a remote command and control center, allowing border authorities to monitor large and difficult-to-access regions without putting soldiers at risk. • Uses a bio-inspired Grouping Cockroaches Classifier (GCC) to classify human gestures instead of facial features. This approach enables the system to identify border soldiers and suspicious individuals (terrorists or intruders) even when faces are hidden or people are wearing disguises, making the system more effective under challenging conditions.

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				• Introduces an IoT-enabled intelligent surveillance architecture that connects drones, sensors, cameras, and command centers through the Internet. The system supports real-time communication, remote monitoring, and quick decision-making, improving the overall efficiency and reliability of border surveillance. • Evaluates the proposed system using performance metrics such as precision, recall, F1-score, entropy, success rate, and error rate, demonstrating better performance than traditional algorithms like K-Means, Decision Tree, and C4.5. The authors also propose future enhancements by integrating deep learning and mobile applications to improve detection accuracy, scalability, and remote accessibility.

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8.	A Comprehensive Study on Human Detection and Tracking Using Multispectral Thermal Imaging in Low-Light Conditions for Military and Border Security Applications	Elsevier	2026	• Proposes a multispectral human detection system for military and border security using two deep learning architectures, InfraVisNet-L (YOLOv11n backbone) and InfraVisNet-H (RT-DETR-L backbone). The system combines visible (RGB) and thermal images to achieve reliable human detection in low-light, foggy, and camouflaged environments. • Uses multispectral datasets (LLVIP and FLIR-ADAS) containing aligned RGB and thermal image pairs to train and evaluate the proposed models. Thermal imaging enables detection based on human heat signatures, allowing accurate surveillance even in darkness, smoke, and adverse weather conditions where conventional cameras fail. • Integrates real-time multi-object tracking (MOT) with advanced tracking algorithms to continuously track multiple individuals while maintaining consistent identities, even in crowded scenes or during temporary occlusions. The system can also generate automatic alerts whenever a person enters a restricted or sensitive area.

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				• Addresses the limitations of traditional surveillance systems by combining thermal imaging with deep learning, reducing false detections caused by poor illumination, camouflage, and visual clutter. This significantly improves human detection accuracy and situational awareness in challenging border environments. • -Develops a practical real-time surveillance framework specifically for military and border security applications. The proposed solution enhances border monitoring by providing accurate human detection, reliable tracking, and rapid threat identification, making it suitable for deployment in difficult terrains and extreme environmental conditions.

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9.	A Human Intruder Detection System for Restricted Sensitive Areas	IEEE	2021	• Proposes a real-time video-based human intrusion detection system for border and sensitive areas using the YOLO (You Only Look Once) object detection model. The system is designed to automatically detect unauthorized human entry and reduce the need for continuous manual surveillance. • Uses low-cost surveillance cameras to capture live video streams, which are processed in real time. The system identifies human intruders and immediately sends alerts to the control room, enabling faster response to security threats. • Employs the YOLO deep learning algorithm for accurate human detection while distinguishing humans from other objects. The proposed model achieves an intrusion detection accuracy of approximately 96% under the tested environmental conditions. • Optimizes storage requirements by automatically deleting recorded video segments that do not contain human intrusion after a predefined time interval. This reduces memory usage and eliminates the need to store unnecessary surveillance footage.

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				• Provides a cost-effective and scalable surveillance solution compared to traditional hardware- and radar-based systems. The authors suggest further enhancement by testing the model in real-time deployment scenarios to improve its robustness and practical implementation for border security applications.

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10.	Development of a Framework for Effective Surveillance and Human Intrusion Detection in Border regions covered with Dry leaves	IEEE	2020	• Proposes an integrated multi-sensor border surveillance system as part of the PANCHENDRIYA framework for detecting human intrusion in border regions covered with dry leaves. The system combines Microphone sensors, Camera sensors, and Wireless Sensor Networks (WSNs) to provide continuous monitoring while reducing manpower and the cost of conventional border patrolling. • Implements microphone-based footstep detection to identify and track human movement through the sound generated by footsteps on dry leaves. The collected acoustic signals are analyzed using polynomial regression, and multiple predefined thresholds are established to improve the accuracy of human intrusion detection. • Integrates camera-based surveillance with the microphone sensor to provide visual confirmation of intrusions. The camera detects the face and movement of the intruder, reducing false alarms and enhancing the reliability of the overall surveillance system.

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				• Designs and validates the intrusion detection system through testbed experiments conducted in a dry-leaf environment. The developed microphone detection circuit achieves very low RMSE values, demonstrating the accuracy and efficiency of the proposed acoustic sensing approach. • Demonstrates an efficient and cost-effective border monitoring solution by combining acoustic sensing, video surveillance, and WSN technology. The authors propose real-time deployment with advanced wireless communication technologies as future work to improve monitoring, remote control, and practical implementation in actual border environments.

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11.	Gait-Based Person Identification and Intruder Detection Using mm-Wave Sensing in Multi-Person Scenario	IEEE	2024	• To identify people and detect intruders using mm-Wave (Millimeter-Wave) radar sensing • To recognize a person's walking style (gait) without using cameras. • To improve security in places where multiple people are present. • Developed a person identification and intruder detection system using mm-Wave radar. • Collected gait (walking pattern) data from multiple people. • Applied deep learning to identify authorized and unauthorized persons. • Tested the system in multi-person environments where several people were moving together. • Evaluated the system for identification accuracy and intrusion detection performance. • mm-Wave radar can identify people without capturing facial images, protecting privacy. • The system accurately recognized people based on their walking pattern.

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				• It successfully detected unauthorized intruders even when multiple people were present. • The system worked well in low-light and nighttime conditions, unlike normal cameras. • Deep learning improved the accuracy and reliability of person identification. • The study shows that mm-Wave sensing is a promising technology for smart surveillance and perimeter security.

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12.	Revolutionizing Perimeter Intrusion Detection: A Machine Learning-Driven Approach	IEEE	2022	• To improve the accuracy of Perimeter Intrusion Detection Systems (PIDS). • To reduce false intrusion alarms. • To detect and classify intrusion activities using machine learning. • To develop a new dataset (STPID-dataset) for future research. • Developed a Perimeter Intrusion Detection System (PIDS) prototype. • Used DHT22, SW-420 vibration sensor, PIR motion sensor, and Arduino UNO to collect intrusion data. • Applied DBSCAN and K-NN algorithms for intrusion detection. • Improved DBSCAN by proposing ST-DBSCAN using automatic epsilon estimation and Manhattan distance. • Created a new STPID-dataset from security camera data. • Tested and compared the proposed system with the existing DBSCAN algorithm. • The system achieved 94.9853% accuracy.

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13.	Intelligent Video Surveillance: A Review Through Deep Learning Techniques	MDPI	2023	• Intelligent Video Surveillance: A Review Through Deep Learning Techniques — MDPI — 2023 • To review how deep learning is used in intelligent video surveillance. • To explain different surveillance techniques and AI models. • To discuss the challenges and future improvements in video surveillance systems. • Reviewed different deep learning models used in video surveillance. • Explained techniques for person detection, object detection, tracking, and activity recognition. • Compared different surveillance methods and public datasets. • Discussed the advantages and limitations of existing deep learning approaches. • Suggested future improvements for intelligent surveillance systems. • Deep learning improves the accuracy of video surveillance systems.

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				• AI can automatically detect and track people and suspicious activities. • Performance is affected by poor lighting, weather conditions, and crowded scenes. • Large and high-quality datasets improve detection performance. • Intelligent surveillance is useful for smart cities, public safety, traffic monitoring, and perimeter security. • Future systems should focus on higher accuracy, faster processing, and real-time monitoring.

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14.	Perimeter Security Utilizing Thermal Object Detection	MDPI	2025	• To develop a thermal camera-based perimeter security system. • To detect intruders using AI object detection models. • To provide 24-hour surveillance in day and night conditions. • Used thermal/infrared (IR) cameras instead of normal RGB cameras. • Collected and combined publicly available thermal datasets. • Removed irrelevant classes and manually labeled missing object annotations. • Evaluated several state-of-the-art deep learning object detection models. • Compared the models based on accuracy (mAP) and real-time processing speed. • Tested the models on mid-range GPUs for practical deployment. • Thermal cameras provide 24-hour monitoring and work well in low-light conditions. • All tested models achieved real-time detection. • The highest detection accuracy was 54.4% mAP.

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				• Larger models and higher image resolutions did not always improve performance. • Medium-sized models offered the best balance between speed and accuracy. • The study concluded that thermal object detection improves perimeter security, but larger and more diverse thermal datasets are needed for better accuracy.

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15.	A Micro Controller Based Monitoring System for Cattle Farm Security	IEEE	2020	• To develop an automated cattle farm security system using a microcontroller. • To protect cattle from theft, intrusion, and unexpected movement. • To continuously monitor the farm area without depending on manual observation. • To provide alerts when unusual activity is detected. • Developed a microcontroller-based monitoring system for cattle farm security. • Used sensors to monitor the farm environment and animal movement. • Integrated sensors with a microcontroller for real-time data processing. • Designed an alert system to notify the owner during abnormal events. • Implemented automatic monitoring to reduce human effort. • Tested the system for detecting unauthorized entry and farm safety conditions. • Microcontroller Unit (MCU): Processes sensor data and controls the system.

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				• PIR Motion Sensor: Detects human or animal movement. • Ultrasonic Sensor: Detects nearby objects by measuring distance. • Camera Module: Captures images/videos for monitoring. • Communication Module: Sends alerts to the user. • Buzzer/Alarm: Provides warning during intrusion. • The system provides real-time monitoring, reduces human effort, and improves cattle farm security by detecting unauthorized activities.

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16.	Improved Change Detector Using Dual-Camera Sensors for Intelligent Surveillance Systems	IEEE	2020	• Introduces a smarter approach to camera-based surveillance using two cameras: a regular color camera and an infrared (IR) camera. • Developed specifically for sterile zone monitoring (SZM) — detecting any object entering a prohibited area, such as a prison fence, industrial zone, or restricted rooftop. • The color camera is used during the day, while the system switches to the IR camera at night. • A Statistical Decision Criterion (SDC) automatically determines the time of day by analyzing patterns in the image, and switches which camera feed is trusted accordingly. • IR footage is specifically enhanced, followed by background subtraction using a Gaussian Mixture Model (GMM) and basic image processing for cleanup. • Validation: the proposed algorithm (ICD) was tested on three separate datasets and outperformed other top-ranked change detection algorithms.

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				• Main transferable idea: the novel dual-camera day/night switching approach, which allows other camera-based anomaly detection systems to reliably monitor across varying lighting conditions. • The authors note potential future integration with deep learning-based intruder classification, which this paper does not yet include.

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17.	Unmanned Aerial Surveillance and Tracking System in Forest Areas for Poachers and Wildlife	IEEE	2024	• Addresses poaching in forest areas using a two-stage detection system: a ground-based wireless sensor network for initial detection, followed by a drone for visual confirmation. • The ground layer consists of multiple nodes with PIC microcontrollers plus sound and PIR sensors, relaying signals until an alert reaches the base station. • Multi-hop routing (AODV) is used to relay signals between nodes, overcoming the ~60m range limitation of the ZigBee protocol used. • The drone navigates autonomously to the triggered node's location using a two-fold collision avoidance system, combining ultrasonic sensors (short-range) and a stereo camera (depth perception) to avoid obstacles in dense forest cover. • Once triggered, the drone visually confirms whether the activity is a genuine threat using a custom-trained YOLO model, achieving a mean Average Precision (mAP) of 0.914 and an F1 score of 0.88. • Hardware prototype used a hexacopter with a Raspberry Pi 4B onboard companion computer, powered by a 2200mAh battery.

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				• Main drawback: low battery life, giving the drone only 7 minutes of flight time. • Provides an innovative implementation of ground surveillance over a large area through multi-node signal routing. • False positive handling occurs at a later stage (drone confirmation), rather than at the initial detection layer. • Ground-level detection is low cost, while the confirmation step on the drone uses heavy computation — keeping overall cost low, since expensive processing only runs occasionally for confirmation.

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18.	Edge AI in Sustainable Farming: Deep Learning-Driven IoT Framework to Safeguard Crops From Wildlife Threats	IEEE	2024	• Addresses wildlife intrusion on farms through a combination of an IoT trigger and an AI detection model. • A laser detection system acts as a cheap, low-power trigger to wake a dormant AI detection system built around a camera. • The camera runs a custom lightweight model, EvoNet, specifically designed for tiny, resource-constrained hardware. • The system comprises both a static component (stationary detection) and a dynamic component (an intelligent rover for tracking and physical deterrence). • Deterrence is achieved via the rover, which provides real-time monitoring and feedback to farmers while physically intervening against wildlife. • The system is power- and cost-conservative, since the smart detection layer only activates upon a signal from the low-level laser trigger. • Results: EvoNet achieved the highest accuracy at 96.7% among the models compared in the paper, and was compressed via pruning and quantization to a file size as small as 1.63MB while maintaining comparable accuracy — making it deployable on constrained edge hardware.

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				• Main drawback: lack of accountability for false positives — the trigger may not be smart enough to filter out noise, potentially waking the dormant AI system far more often than necessary.

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19.	Enhanced IoT-Based Smart Agro-Farm Security System Through Multi-Sensor Fusion and Field Validation	Elsevier	2025	• Presents a real-world, hardcoded security system using six sensors: laser, vibration, PIR motion, pyroelectric motion, MQ-2 gas, and flame/fire sensors. • This is an upgrade of a previous version that relied on only laser and vibration sensors, achieving 90% detection efficiency without accounting for false alarms. • Designed to provide a complete farm safety system, covering both intrusion and imminent hazards (fire, gas leaks) together. • Architecture consists of a sender station (Arduino Mega, collects sensor data) and a receiver station (Arduino Uno, receives alerts), working together to deliver emergency notifications via GSM (SMS-style alerts over cellular networks) — useful in rural areas without reliable WiFi. • Provides strong real-world field deployment data, with accurate and honest sensor performance numbers, valuable for benchmarking sensors in other projects. • Results: the laser sensor achieved 100% detection efficiency across all conditions; overall system efficiency reached 92.5% daytime and 94.38% nighttime; wireless (HC-12) transmission reliability was 98.73%, and GSM notification success was 96.78%.

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				• Testing was conducted both in a controlled lab and in real farm sheds (machinery and dairy sheds), across both day and night conditions. • Limitations are explicitly stated, including a 15% daytime false alarm rate, PIR sensors performing worse during the day than at night, a need for manual laser calibration, reduced reliability in open/obstructed areas, and no weatherproofing, limiting performance in harsh outdoor conditions. • Implements no AI/ML layer at all — full functionality is achieved through hardcoded sensor logic, demonstrating that solid false-alarm reduction is achievable through good sensor engineering alone.

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20.	Towards Preventing the False Alarms in Indoor Physical Intrusion Detector System and the Incorporation of Intruder Immobilizer System	Elsevier	2025	• Presents a low-cost indoor intrusion detection system that not only triggers an alarm but physically immobilizes intruders using a lachrymatory agent (tear gas) as its main deterrent. • Designed specifically to be affordable and reliable for low-income regions of Sub-Saharan Africa, where advanced infrastructure and repair facilities are limited. • Uses two ultrasonic sensors positioned apart, combined via an AND logic gate, requiring both sensors to trigger simultaneously to register an intrusion — a simple hardware-based method to prevent false positives. • Ultrasonic sensors are chosen specifically for having the lowest false alarm rate compared to other sensor options. • Once a trigger is registered, a switch releases gas from a canister containing a lachrymatory agent, incapacitating the intruder. • An alarm is also triggered to warn others, though no digital alert is sent, as this was found unreliable in the target deployment regions (unstable network areas). • Uses an aggressive, hardcoded-logic-only approach with zero AI/ML involved in detection or confirmation. • Keeps both physical and computational cost significantly low, offering useful basic home safety against intrusion and burglary.